

Nail Kocabay

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SUMMARY

Computer Engineering student at Marmara University with fourth-year coursework completed and an 18-day mandatory internship remaining. Experienced in AI/ML, computer vision, backend, and full-stack development.

EXPERIENCE

AI Engineering Intern

September 2025 – May 2026

Payantis Finansal Teknoloji A.Ş., İstanbul

Developed fraud detection pipelines across **16M+ transactions**, including preprocessing, feature engineering, card profiling, and fraud reasoning.

Engineered **13 behavioral features** and applied **MiniBatchKMeans**, **RAPIDS cuML KMeans**, and GPU-accelerated processing for card behavior analysis.

Trained an **XGBoost model** achieving **99.56% fraud recall**, detecting **4,775 of 4,796 fraudulent transactions** on a **5M+ transaction test set**.

Built a multi-agent AI fraud reasoning system using **Pandas**, **Dask**, **parallel processing**, and **LLM-based analysis**.

Engineering Intern

July 2025 – September 2025

JMW Jant Sanayi A.Ş., Aydın

Worked on **Windows Server administration**, **network security**, **system configuration**, and **enterprise IT infrastructure**.

Intern

August 2024 – September 2024

FLO Mağazacılık A.Ş., İstanbul

Developed responsive web components using **HTML5** and **CSS3**, applying semantic HTML and responsive design techniques.

PROJECTS

Okey Game Analysis – Computer Vision and Optimization

Developed an Okey tile detection and classification pipeline using **OpenCV**, **YOLOv8-Nano**, and **ResNet-18**.

Built a dataset of **219 images**, **4,432 annotations**, and **55 classes**; achieved **99.4% mAP50** and **97.26% classification accuracy**.

Implemented **DFS with memoization** for tile-hand optimization, achieving sub-**10 ms** optimization and approximately **200 ms end-to-end** response time.

Integrated the system into a web application using **Next.js**, **TypeScript**, **Flask**, and **REST APIs**.

AI-Powered Product Image Generation System

Developed a local GPU-based generative AI application that transforms one product image into **front**, **side**, and **isometric views**.

Integrated **Qwen Image Edit**, **ComfyUI**, **VAE**, **CLIP**, **UNET**, **Lightning LoRA**, and **AuraFlow**.

Built a **React 18** and **Vite** frontend with **WebSocket-based progress tracking**; achieved approximately **30–60 seconds** per product without paid cloud APIs.

Multi-Agent Fraud Reasoning System

Designed specialized agents for data analysis, financial reasoning, legal analysis, and final decision making using transaction history, behavioral features, and card-level clustering.

Achieved **71.43% precision**, **100% recall**, **83.33% binary F1**, and **79.17% macro F1**.

Spotify Music Recommendation Platform

Developed a content-based recommendation platform using **Python**, **FastAPI**, **scikit-learn**, **Pandas**, and **NumPy**.

Implemented a **KNN engine** with cosine distance, feature scaling, and **10 audio features**; cleaned **114K+ tracks** to **89,740 usable records**.

TECHNICAL SKILLS

Languages: Python, C#, Java, C, SQL, JavaScript, TypeScript, HTML5, CSS3

AI/ML: Machine Learning, Deep Learning, Fraud Detection, Feature Engineering

Computer Vision: YOLOv8, ResNet, OpenCV, Object Detection, Image Classification, Image Processing

Generative AI: LLMs, Prompt Engineering, Qwen Image Edit, ComfyUI, Hugging Face, VAE, CLIP, LoRA

Frameworks: ASP.NET Core, React, Next.js, FastAPI, Flask, scikit-learn, PyTorch, TensorFlow, Pandas, NumPy

Tools: REST APIs, WebSocket, SQL Server, PostgreSQL, Git, Docker, Azure, GCP, Linux

EDUCATION

Marmara University

Istanbul, Türkiye

Computer Engineering, 83.8 / 100

2022 – 2026

LANGUAGE

Turkish: Native

English: B2